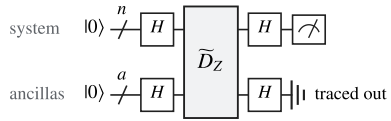
(a) Ancilla-extended IQP circuit



$\widetilde{D}_Z$ in the ancilla basis

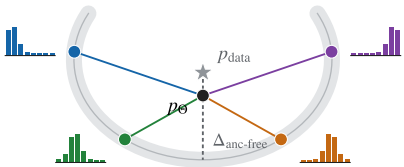
ℓ: 00 01 10 11

$$\begin{bmatrix} \theta^{(0)} & & & \\ & \theta^{(1)} & & \\ & & \theta^{(2)} & \\ & & & \theta^{(3)} \end{bmatrix}$$

$$\theta_G^{(\ell)} = \sum_S (-1)^{S\cdot\ell}\,\widetilde{\theta}_{G,S}$$

(b) Beating a single circuit needs diversity

$$p_\Theta = \sum_\ell \pi_\ell\, p_{\theta^{(\ell)}}$$



single ancilla-free circuits

gain requires $D_{\text{branch}} \geq \Delta_{\text{anc-free}} - \sqrt{\mathcal{L}}$

(c) Which branch differences help

data-agnostic or global start:
$D_{\text{branch}} = 0$, no separating gradient,
GD escape time $\propto \ln(1/\delta)$

diverse, not aligned:
$D_{\text{branch}} > 0$, $\Lambda \approx 1$,
removing a branch helps

aligned with the target:
$\Lambda \gg 1$, each branch covers
a part the others miss